Schroder et al., 1990. See above.

*This gives a thorough discussion of conductivity measurements as they are applied to semiconductors (four-point probing, Van der Pauw methods).*

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# HALL EFFECT IN SEMICONDUCTORS

## INTRODUCTION

The Hall effect, which was discovered in 1879 (Hall, 1879), determines the concentration and type (negative or positive) of charge carriers in metals, semiconductors, or insulators. In general, the method is used in conjunction with a conductivity measurement to also determine the mobility (ease of movement) of the charge carriers. At low temperatures and high magnetic fields, quantum effects are sometimes evident in lower dimensional structures; however, such effects will not be considered here. Also, this unit will concentrate on semiconductors, rather than metals or insulators, although the same theory generally applies (Gantmakker and Levinson, 1987).

Three, strong advantages of Hall effect measurements are ease of instrumentation, ease of interpretation, and wide dynamic range. With respect to implementation, the only elements necessary, at the lowest level, are a current source, a voltmeter, and a modest-sized magnet. The carrier concentration can then be calculated within a typical accuracy of 20% without any other information about the material. In our laboratory, we have measured concentrations ranging from $10^4$ to $10^{20}$ $cm^{-3}$. Also, the type ($n$ or $p$) can be unambiguously determined from the sign of the Hall voltage.

Competing techniques include capacitance-voltage ($C$-$V$) measurements to determine carrier concentration (see CAPACITANCE–VOLTAGE (C-V) CHARACTERIZATION OF SEMICONDUCTORS); thermoelectric probe (TEP) measurements to determine carrier type; and magnetoresistance (MR) measurements to determine mobility (Look, 1989). The $C$-$V$ measurements have an advantage of depth profile information but a disadvantage of requiring a Schottky barrier. The TEP measurements require a temperature gradient to be imposed and are often ambiguous in the final conclusions. The MR measurements require a high mobility $\mu$ or high magnetic field strength $B$ to implement, because the signal varies as $\mu^2B^2$ rather than $\mu B$, as in the Hall effect. Further comparisons of these techniques, as well as others, can be found in monographs by Runyan (1975), Look (1989), and Schroder (1990).

## PRINCIPLES OF THE METHOD

A phenomenological equation of motion for electrons of charge $-e$ moving with velocity $\mathbf{v}$ in the presence of electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ is

$$m^*\dot{\mathbf{v}} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) - m^*\frac{\mathbf{v} - \mathbf{v}_{eq}}{\tau} \tag{1}$$

where $m^*$ is the effective mass, $\mathbf{v}_{eq}$ is the velocity at equilibrium (steady state), and $\tau$ is the velocity or (momentum) relaxation time, i.e., the time in which oscillatory phase information is lost through collisions. Consider a rectangular sample, as shown in Figure 1, with an external electric field $\mathbf{E}_{ex} = E_x\mathbf{x}$ and magnetic field $\mathbf{B} = B_z\mathbf{z}$. Then, if no current is allowed to flow in the $y$ direction (i.e., $v_y = 0$), the steady-state condition $\dot{\mathbf{v}} = 0$ requires that $E_y = -v_xB_z$, and $E_y$ is known as the Hall field. For electron concentration $n$, the current density $j_x = nev_x$, and thus, $E_y = -j_xB_z/en \equiv -j_xB_zR_H$, where $R_H = -1/en$ is the Hall coefficient. Thus, simple measurements of the quantities $E_y$, $j_x$, and $B_z$ yield a very important quantity, $n$, although a more detailed analysis given below slightly modifies this relationship.

The above analysis assumes that all electrons are moving with the same velocity $\mathbf{v}$ (constant $\tau$), which is not true in a semiconductor. To relieve this constraint, we note that Equation 1 consists of three, coupled differential equations (in $v_x$, $v_y$, $v_z$) that can be solved by standard techniques. After averaging over energy, the steady-state currents can then be shown to be

$$j_x = \sigma_{xx}E + \sigma_{xy}E_y \tag{2}$$

$$j_y = \sigma_{yx}E_x + \sigma_{yy}E_y \tag{3}$$

where the $\sigma_{ij}$ are elements of the conductivity tensor, defined by

$$\sigma_{xx} = \sigma_{yy} = \left\langle \frac{\tau}{1 + \omega_c^2\tau^2} \right\rangle \tag{4}$$

$$\sigma_{xy} = -\sigma_{yx} = -\left\langle \frac{\omega_c\tau^2}{1 + \omega_c^2\tau^2} \right\rangle \tag{5}$$

Here $\omega_c = eB/m^*$ is the cyclotronic frequency, where $B$ is the magnitude of $\mathbf{B}$, and the brackets denote an average over energy $\varepsilon$ taken as follows:

$$\langle F(\varepsilon)\rangle = \frac{\int_0^\infty F(\varepsilon)\varepsilon^{3/2}\frac{\partial f_0}{\partial \varepsilon}d\varepsilon}{\int_0^\infty \varepsilon^{3/2}\frac{\partial f_0}{\partial \varepsilon}d\varepsilon} \rightarrow \frac{\int_0^\infty F(\varepsilon)\varepsilon^{3/2}e^{-\varepsilon/kT}d\varepsilon}{\int_0^\infty \varepsilon^{3/2}e^{-\varepsilon/kT}d\varepsilon} \tag{6}$$

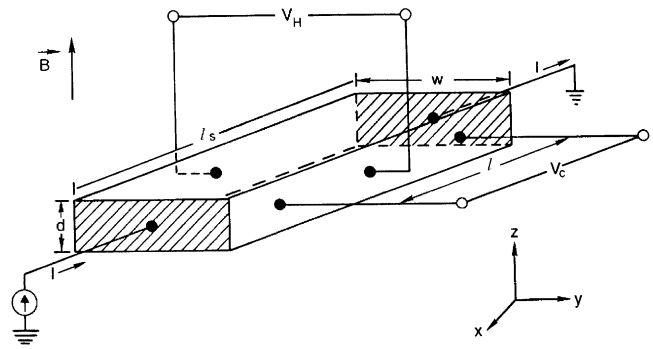



**Figure 1.** Hall bar configuration for resistivity and Hall effect measurements.

where $F(\varepsilon)$ is any function of $\varepsilon$ and $f_0$ is the Fermi-Dirac distribution function. The second equality in Equation 6 holds for nondegenerate electrons, i.e., those describable by a Boltzmann distribution function. For small magnetic fields, i.e., $\omega_c\tau \ll 1$, and under the usual constraint $j_y=0$, it is easy to show that

$$j_x = \frac{ne^2\langle\tau\rangle}{m^*}E_x \equiv -ne\mu_c E_x \tag{7}$$

$$R_H = \frac{E_y}{j_x B} = -\frac{1}{ne}\frac{\langle\tau^2\rangle}{\langle\tau\rangle^2} = -\frac{r}{en} \tag{8}$$

where $r$ is the "Hall factor" and $\mu_c = e\langle\tau\rangle/m^*$ is known as the "conductivity" mobility, since the quantity $ne\mu_c$ is just the conductivity $\sigma$. We define the "Hall" mobility as $\mu_H = R_H\sigma = r\mu_c$ and the "Hall" concentration as $n_H = n/r = -1/eR_H$. Thus, a combined Hall effect and conductivity measurement gives $n_H$ and $\mu_H$, not $n$ and $\mu_c$; fortunately, however, $r$ is usually within 20% of unity and almost never as large as 2. In any case, $r$ can often be calculated or measured $[r = R_H(\mu_B \ll 1)/R_H(\mu_B \gg 1)]$ so that an accurate value of $n$ can usually be determined.

The relaxation time $\tau(\varepsilon)$ depends on how the electrons interact with the lattice vibrations as well as with extrinsic elements, such as charged impurities and defects. For example, acoustical-mode lattice vibrations scatter electrons through the deformation potential (leading to a relaxation time $\tau_{ac}$) and piezoelectric potential ($\tau_{pe}$); optical-mode vibrations through the polar potential ($\tau_{po}$); ionized impurities and defects through the screened coulomb potential ($\tau_{ii}$); and charged dislocations, also through the coulomb potential ($\tau_{dis}$). The strengths of these various scattering mechanisms depend upon certain lattice parameters, such as dielectric constants and deformation potentials, and extrinsic factors, such as donor, acceptor, and dislocation concentrations $N_D$, $N_A$, and $N_{dis}$, respectively (Rode, 1975; Wiley, 1975; Nag, 1980; Look, 1989). The total momentum scattering rate, or inverse relaxation time, is

$$\tau^{-1}(\varepsilon) = \tau_{ac}^{-1}(\varepsilon) + \tau_{pe}^{-1}(\varepsilon) + \tau_{po}^{-1}(\varepsilon) + \tau_{ii}^{-1}(\varepsilon) + \tau_{dis}^{-1}(\varepsilon) \tag{9}$$

This expression is then used to determine $\langle\tau(\varepsilon)\rangle$ and $\langle\tau^2(\varepsilon)\rangle$ via Equation 6, and thence, $\mu_H = e\langle\tau^2\rangle/m^*\langle\tau\rangle$. Since $\mu_H$ is a function of temperature $T$, a fit of $\mu_H$ vs. $T$ can often be used to determine $N_D$, $N_A$, and $N_{dis}$. The $\mu_H$-vs.-$T$ curve should be solved simultaneously with the $n$-vs.-$T$ curve, given by the charge-balance equation (CBE)

$$n + N_A = \frac{N_D}{1 + n/\phi_D} \tag{10a}$$

where $\phi_D = (g_0/g_1)N_C' \exp(\alpha_D/k)T^{3/2}\exp(-E_{D0}/kT)$. Here, $g_0/g_1$ is a degeneracy factor ($= 1/2$ for an $s$-state), $N_C' = 2(2\pi m_n^* k)^{3/2}/h^3$, where $h$ is Planck's constant, $E_D$ is the donor energy, $k$ is Boltzmann's constant, and $E_{D0}$ and $\alpha_D$ are defined by $E_D = E_{D0} - \alpha_D T$. If more than one donor exists within a few $kT$ of the Fermi energy, then equivalent terms are added on the right-hand side of Equation 10a.

For a $p$-type sample, we use the nearly equivalent equation

$$p + N_D = \frac{N_A}{1 + p/\phi_A} \tag{10b}$$

where $p$ is the hole concentration and $\phi_A = (g_1/g_0)N_V' \exp(\alpha_A/k)T^{3/2}\exp(-E_{A0}/kT)$, and where $N_V' = 2(2\pi m_p^* k)^{3/2}/h^3$ and $E_A = E_{A0} - \alpha_A T$.

The mobility analysis described above, known as the relaxation time approximation (RTA), is limited to elastic (energy-conserving) scattering processes, because for these, a relaxation time $\tau(\varepsilon)$ can be well defined. Unfortunately, this is usually not the case for polar optical-mode (po) scattering, since the energy exchanged in the po scattering event is typically $\gg kT$. However, we can often approximate $\tau_{po}$ by an analytic formula, and, in any case, at low temperatures, po scattering is not very important. Nevertheless, for the most accurate calculations, the Boltzmann transport equation (BTE) should be solved directly, as discussed in several references (Rode, 1975; Nag, 1980; Look, 1989).

Hall samples do not have to be rectangular, such as that shown in Figure 1; in fact, we will discuss arbitrarily shaped specimens below (see Practical Aspects of the Method). However, the above analysis does assume that $n$ and $\mu$ are homogeneous throughout the sample. If $n$ and $\mu$ vary with depth $z$ only, then the measured quantities are

$$\sigma_{sq} = \int_0^d \sigma(z)\,dz = e\int_0^d n(z)\mu(z)\,dz \tag{11}$$

$$R_{Hsq}\sigma_{sq}^2 = \int_0^d n(z)\mu^2(z)\,dz \tag{12}$$

where $d$ is the sample thickness and the subscript "sq" denotes a sheet (areal) quantity (in reciprocal centimeters squared) rather than a volume quantity (in reciprocal cubic centimeters). If some of the carriers are holes rather than electrons, then the sign of $e$ for those carriers must be reversed. The general convention is that $R_H$ is negative for electrons and positive for holes. In some cases, the hole and electron contributions to $R_{Hsq}^2\sigma_{sq}$ exactly balance at a given temperature, and this quantity vanishes.

## PRACTICAL ASPECTS OF THE METHOD

### Useful Equations

The Hall bar structure of Figure 1 is analyzed as follows: $E_x = V_c/l$, $E_y = V_H/w$, and $j_x = I/wd$, so that

$$\sigma = \rho^{-1} = \frac{j_x}{E_x} = \frac{Il}{V_c wd} \tag{13}$$

$$R_H = \frac{E_y}{j_x B} = \frac{V_H d}{IB} \tag{14}$$

$$\mu_H = R\sigma = \frac{V_H l}{V_c wB} \tag{15}$$

$$n_H = (eR)^{-1} \tag{16}$$

In the meter-kilogram-second (mks) system, current $I$ is in amperes, voltage $V$ in volts, magnetic field strength $B$ in tesla, and length $l$, width $w$, and thickness $d$ in meters. By realizing that $1\ \mathrm{T} = 1\ \mathrm{V \cdot s/m^2}$, $1\ \mathrm{A} = 1\ \mathrm{C/s}$, and $1\ \Omega = 1\ \mathrm{V/A}$, we find that $\sigma$ is in units of $\Omega^{-1}\mathrm{m}^{-1}$, $R_\mathrm{H}$ in $\mathrm{m^3/C}$, $\mu_\mathrm{H}$ in $\mathrm{m^2/V \cdot s}$, and $n_\mathrm{H}$ in $\mathrm{m^{-3}}$. However, it is more common to denote $\sigma$ in $\Omega^{-1}\ \mathrm{cm^{-1}}$, $R_\mathrm{H}$ in $\mathrm{cm^3/C}$, $\mu_\mathrm{H}$ in $\mathrm{cm^2/V \cdot s}$, and $n_\mathrm{H}$ in $\mathrm{cm^{-3}}$, with obvious conversion factors ($1\ \mathrm{m} = 10^2\ \mathrm{cm}$). Since $B$ is often quoted in gauss, it is useful to note that $1\ \mathrm{T} = 10^4\ \mathrm{G}$.

Although the Hall bar configuration discussed above is the simplest and most straightforward geometry to analyze, it is not the most popular shape in use today. The reason stems from a very convenient formulation by van der Pauw (1958) in which he solved the potential problem for a thin layer of arbitrary shape. A convenient feature of the van der Pauw technique is that no dimension need be measured for the calculation of sheet resistance or sheet carrier concentration, although a thickness must of course be known for volume resistivity and concentration. Basically, the validity of the van der Pauw method requires that the sample be flat, homogeneous, isotropic, and a singly connected domain (no holes) and have line electrodes on the periphery, projecting to point contacts on the surface, or have true point contacts on the surface. The last requirement is the most difficult to satisfy, so that much work has gone into determining the effects of finite contact size.

Consider the sample shown in Figure 2A. Here, a current $I$ flows between contacts 1 and 2, and a voltage $V_\mathrm{c}$ is measured between contacts 3 and 4. Let resistance $R_{ij,kl} \equiv V_\mathrm{kl}/I_\mathrm{ij}$, where the current enters contact $i$ and leaves contact $j$ and $V_{kl} = V_k - V_l$. [These definitions, as well as the contact numbering, correspond to ASTM (1988) Standard F76.] The resistivity $\rho$, with $B = 0$, is then calculated as

$$\rho = \frac{\pi d}{\ln(2)}\left(\frac{R_{21,34} + R_{32,41}}{2}\right) f \tag{17}$$

where $f$ is determined from a transcendental equation:

$$\frac{Q-1}{Q+1} = \frac{f}{\ln(2)}\operatorname{arccosh}\left[\frac{1}{2}\exp\left(\frac{\ln(2)}{f}\right)\right] \tag{18}$$

Here, $Q = R_{21,34}/R_{32,41}$ if this ratio is greater than unity; otherwise, $Q = R_{32,41}/R_{21,34}$. A curve of $f$ vs. $Q$ accurate to $\approx 2\%$ is presented in Figure 3 (van der Pauw, 1958). Also useful is a somewhat simpler analytical procedure

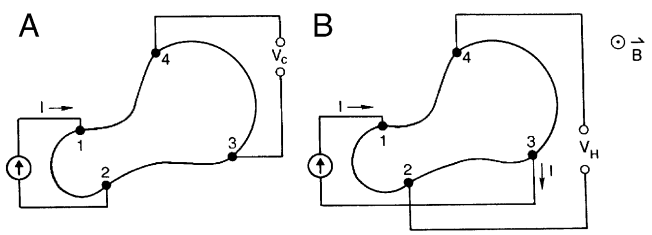



**Figure 2.** Arbitrary shape for van der Pauw measurements: (**A**) resistivity; (**B**) Hall effect.

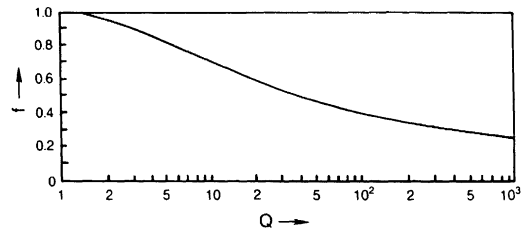



**Figure 3.** Resistivity ratio function used to correct the van der Pauw results for asymmetric sample shape.

for determining $f$ due to Wasscher and reprinted in Weider (1979). First calculate $\alpha$ from

$$Q = \frac{\ln(1/2 - \alpha)}{\ln(1/2 + \alpha)} \tag{19}$$

and then calculate $f$ from

$$f = \frac{\ln(1/4)}{\ln(1/2 + \alpha) + \ln(1/2 - \alpha)} \tag{20}$$

Here, it is of course required that $-1/2 < \alpha < 1/2$, but this range of $\alpha$ covers $Q = 0$ to $Q = \infty$. For example, a ratio $Q = 4.8$ gives a value $\alpha \approx 0.25$, and then $f \approx 0.83$. Thus, the ratio must be fairly large before $\rho$ is appreciably reduced.

It is useful to further average $\rho$ by including the remaining two contact permutations, and also reversing current for all four permutations. Then

$$\rho = \frac{\pi d}{\ln(2)}\left[\frac{(R_{21,34} - R_{12,34} + R_{32,41} - R_{23,41})f_\mathrm{A}}{8} + \frac{(R_{43,12} - R_{34,12} + R_{14,23} - R_{41,23})f_\mathrm{B}}{8}\right] \tag{21}$$

where $f_\mathrm{A}$ and $f_\mathrm{B}$ are determined from $Q_\mathrm{A}$ and $Q_\mathrm{B}$, respectively, by applying either Equation 18 or 19. Here,

$$Q_\mathrm{A} = \frac{R_{21,34} - R_{12,34}}{R_{32,41} - R_{23,41}} \tag{22}$$

$$Q_\mathrm{B} = \frac{R_{43,12} - R_{34,12}}{R_{14,23} - R_{41,23}} \tag{23}$$

The Hall mobility is determined by using the configuration of Figure 2B, in which the current and voltage contacts are crossed. The Hall coefficient becomes

$$R_\mathrm{H} = \frac{d}{B}\left(\frac{R_{31,42} + R_{42,13}}{2}\right) \tag{24}$$

In general, to minimize magnetoresistive and other effects, it is useful to average over current and magnetic field polarities. Then

$$R_\mathrm{H} = (d/B)[R_{31,42}(+B) - R_{13,42}(+B) + R_{42,13}(+B) - R_{24,13}(+B) + R_{13,42}(-B) - R_{31,42}(-B) + R_{24,13}(-B) - R_{42,13}(-B)]/8 \tag{25}$$

### Sensitivity

The most common magnetic field strength used by researchers for Hall effect measurements in 5 kG = 0.5 T. Also, typical semiconductor materials (e.g., GaAs) vary in resistivity $\rho$ from $10^{-4}$ to $10^{9}$ $\Omega \cdot$ cm, depending on intrinsic factors as well as impurity and defect concentrations. Finally, typical sample thicknesses $d$ may range from $10^{-5}$ to $10^{-1}$ cm. Consider the worst case on the high-resistivity end of the spectrum. Let $\rho = 10^{9}$ $\Omega \cdot$ cm, $d = 10^{-5}$ cm, and $l/w \approx 1$. Then, if a voltage of 100 V can be applied, the current will be 1 pA, which can be measured with an electrometer. If the mobility $\mu_H \approx 10^{3}$ cm$^2$/V $\cdot$ s, then the Hall coefficient $R$ will be $10^{12}$ cm$^3$/C, and the Hall voltage $V_H$ will be 5 V. Conversely, let $\rho = 10^{-4}$ $\Omega \cdot$ cm and $d = 10^{-1}$ cm. Then, the resistance is about $10^{-3}$ $\Omega$, and we will not be able to apply more than about $10^{-3}$ V in order to avoid sample heating. Such a voltage will produce 1 A of current and a resulting Hall voltage of 50 $\mu$V, again easily measurable. Of course, if $\mu H \approx 1$ cm$^2$/V $\cdot$ s, then $V_H \approx 50$ nV, which can still probably be measured if noise levels are low. We may note that the lowest mobility that we have measured in our laboratory, by using our standard dc techniques, is 0.1 cm$^2$/V s.

## METHOD AUTOMATION

A basic design for a high-impedance, automated van der Pauw–Hall apparatus, which can accommodate samples of up to $10^{12}$ $\Omega$ is given in Figure 4. All components are commercially available, and the electrometers can, of course, be replaced by high-impedance, unity-gain buffer amplifiers, although possibly at some sacrifice in input impedance. The inputs and outputs of the high-impedance scanner and the inputs to the ammeter and electrometers are of triaxial design, which has the advantage that the inner shields can be driven by the unity-gain output of the electrometers, which effectively reduces cable-charging effects. The outer shields are all grounded at a common point, although in practice the grounding may not be critical. The current source should have an effective output impedance of about $10^{12}$ $\Omega$ and be able to regulate currents of $10^{-10}$ A. At higher resistance levels, the current regulation will cross over to voltage regulation, and the currents will diminish; however, the data will still be accurate as long as the actual current and voltage are both measured. The low-impedance scanner should be mainly designed for low thermal offsets (say, a few microvolts) since current leakage is not a problem on this side of the electrometers. Note that although the ammeter needs to be of electrometer design, to be able to measure the very low currents, the voltmeter does not, since the electrometer output impedances are only a few kilohms.

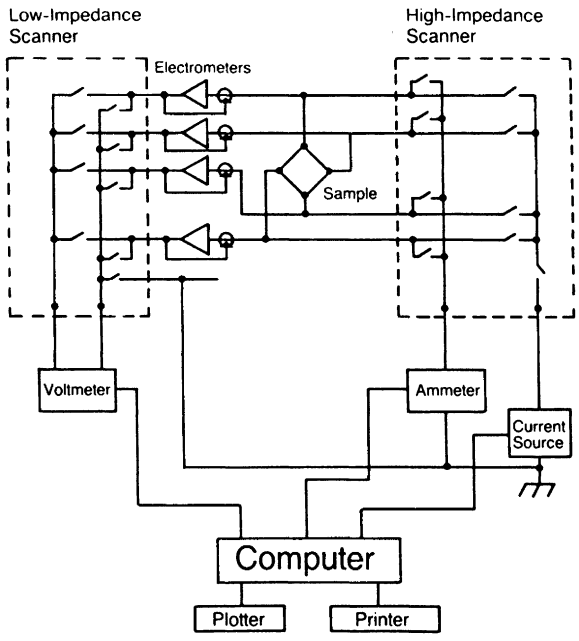



**Figure 4.** Schematic of an automated, high-impedance Hall effect apparatus.

The wiring to the high-impedance scanner in this design allows the current source and current sink to be individually applied to any desired sample contact. Also, the high input and low input on the voltmeter can be connected to any electrometer unity-gain output through the low-impedance scanner. Figure 4 does not represent the most efficient design in terms of the minimum number of scanner switches, but it does illustrate the main ideas quite well. Although not shown here, for very low resistance samples it is desirable, and sometimes necessary, to be able to bypass the electrometers, since their typical noise levels of 10-$\mu$V peak to peak may be unacceptably high.

We will not discuss the computer and peripherals in any detail here, because so many variations available today will work well. However, we would recommend that the system be designed around the standard IEEE-488 interface bus, which allows new instruments to be added at will and effectively eliminates serious hardware problems.

Complete Hall systems, including temperature and magnetic field control, are available from several companies, including Bio-Rad Semiconductor, Keithley Instruments, Lake Shore Cryotronics, and MMR Technologies.

## DATA ANALYSIS AND INITIAL INTERPRETATION

The primary quantities determined from Hall effect and conductivity measurements are the Hall carrier concentration $n_H$ or $p_H$ and mobility $\mu$H. As already discussed, $n_H = -1/eR_H$, where $R_H$ is given by Equation 24 (for a van der Pauw configuration) and $\mu_H = R_H\sigma = R_H/\rho$, where $\rho$ is given by Equation 21. Although simple 300-K values of $\rho$, $n_H$, and $\mu_H$ are quite important and widely used, it is in temperature-dependent Hall (TDH) measurements that the real power of the Hall technique is demonstrated, because then the donor and acceptor concentrations and energies can be determined. We will illustrate the methodology with a two-layer GaN problem.

The GaN sample discussed here was a square (6 $\times$ 6-mm) layer grown on sapphire to a thickness $d = 20$ $\mu$m.

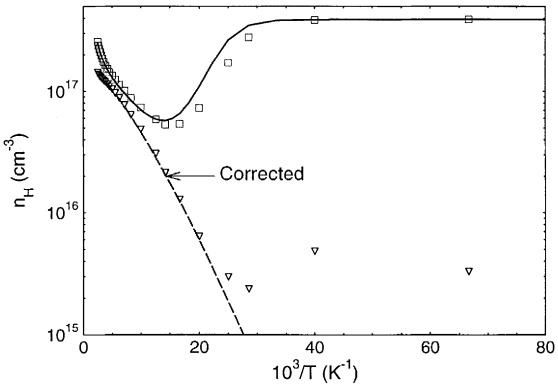



**Figure 5.** Uncorrected Hall concentration data (squares) and fit (solid line) and corrected data (triangles) and fit (dashed line) vs. inverse temperature.

Small indium dots were soldered on the corners to provide ohmic contacts, and the Hall measurements were carried out in an apparatus similar to that illustrated in Figure 4. Temperature control was achieved by using a He exchange-gas dewar. The temperature dependences of $n_H$ and $\mu_H$ are shown by the squares in Figures 5 and 6, respectively. At temperatures below 30 K (i.e., $10^3/T > 33$), the carriers (electrons) in the main part of the layer "freeze out" on their parent donors and thus are no longer available for conduction. However, this sample had a very thin, strongly $n$-type layer between the sapphire and GaN, and the carriers in such a layer do not freeze out and, in fact, have a temperature-independent concentration and mobility, as seen at low $T$ (high $10^3/T$) in Figures 5 and 6, respectively. Thus, we need to use the depth-profile analysis given by Equations 11 and 12.

For two layers, Equations 11 and 12 give

$$\mu_H = R_{sq}\sigma_{sq} = \frac{R_{sq}\sigma_{sq}^2/d}{\sigma_{sq}/d} = \frac{\mu_{H1}^2 n_{H1} + \mu_{H2}^2 n_{Hsq2}/d}{\mu_{H1} n_{H1} + \mu_{H2} n_{Hsq2}/d} \tag{26}$$

$$n_H = \frac{n_{Hsq}}{d} = \frac{1}{eR_{sq}d} = \frac{\sigma_{sq}^2/d^2}{eR_{sq}\sigma_{sq}^2/d} = \frac{(\mu_{H1} n_{H1} + \mu_{H2} n_{Hsq2}/d)^2}{\mu_{H1}^2 n_{H1} + \mu_{H2}^2 n_{Hsq2}/d} \tag{27}$$

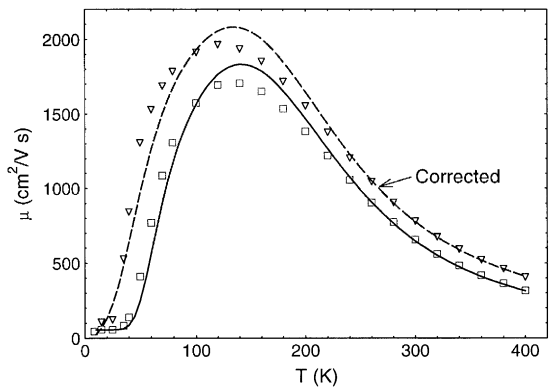



**Figure 6.** Uncorrected Hall mobility data (squares) and fit (solid line) and corrected data (triangles) and fit (dashed line) vs. temperature.

where layer 1 is the main, 20-μm-thick GaN growth and layer 2 is the very thin interface region. Since we do not know the thickness of layer 2, we simply normalize to thickness $d = 20$ μm for plotting purposes. From the figures, we get $\mu_{H2} = 55$ cm$^2$/V · s and $n_{Hsq2}/d = 3.9 \times 10^{17}$ cm$^{-3}$. Because these values are constant, we can invert Equations 26 and 27 and solve for $\mu_{H1}$ and $n_{H1}$ at each temperature. (The resulting equations are given later.) To fit the uncorrected data (squares), we parametrize $n$ vs. $T$ by Equation 10a and $\mu_H$ vs. $T$ by $\mu_H = e\langle\tau^2\rangle/m^*\langle\tau\rangle$, where $\tau$ is given by Equation 9. Because $n = n_H r = n_H\langle\tau^2\rangle/\langle\tau\rangle^2$ the fits of $n_H$ vs. $T$ and $\mu_H$ vs. $T$ must be carried out simultaneously. In this case $r$ varies only from about 1.2 to 1.4 as a function of $T$.

Formulas for $\tau_{ac}, \tau_{pe}, \tau_{po}$, and $\tau_{ii}$ are given below. For ionized impurity (or defect) scattering in a nondegenerate, $n$-type material,

$$\tau_{ii}(\varepsilon) = \frac{2^{9/2}\pi\varepsilon_0^2 (m^*)^{1/2}\varepsilon^{3/2}}{e^4(2N_A + n)[\ln(1+y) - y/(1+y)]} \tag{28}$$

where $y = 8\varepsilon_0 m^* kT\varepsilon/\hbar^2 e^2 n$. Here $\varepsilon_0$ is the low-frequency (static) dielectric constant. [If the sample is $p$ type, let $(2N_A + n) \rightarrow (2N_D + p)$]. For acoustic-mode deformation-potential scattering,

$$\tau_{ac}(\varepsilon) = \frac{\pi\hbar^4\rho_d s^2\varepsilon^{-1/2}}{2^{1/2}E_1^2 (m^*)^{3/2} kT} \tag{29}$$

where $\rho_d$ is the density, $s$ is the speed of sound, and $E_1$ is the deformation potential. For acoustic-mode piezoelectric-potential scattering,

$$\tau_{pe}(\varepsilon) = \frac{2^{2/3}\pi\hbar^2\varepsilon_0\varepsilon^{1/2}}{e^2P^2(m^*)^{1/2}kT} \tag{30}$$

where $P$ is the piezoelectric coupling coefficient [$P = (h_{pz}^2/\rho s^2\varepsilon_0)^{1/2}$]. Finally, for polar optical-mode scattering, a rough approximation can be given

$$\tau_{po}(\varepsilon) = (C_{po}2^{3/2}\pi\hbar^2(e^{T_{po}/T} - 1)[0.762\varepsilon^{1/2} + 0.824(kT_{po})^{1/2} - 0.235(kT_{po})^{-1/2}\varepsilon])/[e^2 kT_{po}(m^*)^{1/2}(\varepsilon_\infty^{-1} - \varepsilon_0^{-1})] \tag{31}$$

where $T_{po}$ is the Debye temperature, $\varepsilon_\infty$ is the high-frequency dielectric constant, and $C_{po}$ is a fitting parameter, of order unity, to correct for the inexact nature of $\tau_{po}(\varepsilon)$. That is, if we had *only* po scattering, then the exact (i.e., BTE) calculation of μH vs. $T$ would be almost identical with the RTA calculation (i.e., $\mu_H = e\langle\tau^2\rangle/m*\langle\tau\rangle$), with $C_{po} = 1$. However, if other scattering mechanisms are also important, then the correction factor $C_{po}$ will be dependent on the relative strengths of these other mechanisms. As an example, to get a good fit to high-temperature (>300 K) $\mu_H$-vs.-$T$ data in GaN (Fig. 6), we use $C_{po} \approx 0.6$. Fortunately, below 150 K, the po mechanism

is no longer important in GaN, and the RTA approach is quite accurate.

The RTA analysis discussed above and the CBE analysis discussed earlier (Equation 10a or 10b) constitute a powerful method for the determination of $N_D$, $N_A$, and $E_D$ in semiconductor material. It can be easily set up on a personal computer, using, e.g., MATHCAD software. In many cases, it is sufficient to simply assume $n = n_H$ (i.e., $r = 1$) in Equations 10a and 28, but a more accurate answer can be obtained by using the following steps: (1) let $n = n_H = 1/eR_H$ at each $T$; (2) fit $\mu_H$ vs. $T$ using $\mu_H = \langle\tau^2\rangle/m^*\langle\tau\rangle$, and get a value for $N_A$; (3) calculate $r = \langle\tau^2\rangle/\langle\tau\rangle^2$ at each $T$; (4) calculate a new $n = rn_H$ at each $T$; and (5) fit $n$ vs. $T$ to Equation 10a and get values of $N_D$ and $E_D$. Further iterations can be carried out, if desired, but usually add little accuracy.

For the benefit of the reader who wishes to set up such an analysis, we give the scattering strength parameters and fitting parameters used for the GaN data fits in Figures 5 and 6. Further discussion can be found in the work of Look and Molnar (1997). From the GaN literature we find: $E_1 = 9.2$ eV $= 1.47 \times 10^{-18}$ J; $P = 0.104$, $\varepsilon_0 = 10.4(8.8542 \times 10^{-12})$ F/m; $\varepsilon_\infty = 5.47(8.8542 \times 10^{-12})$ F/m; $T_{po} = 1044$ K; $m^* = 0.22(9.1095 \times 10^{-31})$ kg; $\rho_d = 6.10 \times 10^3$ kg/m$^3$; $s = 6.59 \times 10^3$ m/s; $g_0 = 1$; $g_1 = 2$; $\alpha_D = 0$; and $N'_C = 4.98 \times 10^{20}$ m$^{-3}$. With all the parameters in mks units, $\mu$H is in units of m$^2$/V$\cdot$ s; a useful conversion is $\mu_H$ (cm$^2$/V$\cdot$ s) $= 10^4$ $\mu_H$ (m$^2$/V $\cdot$ s). To fit the high-$T$ $\mu_H$ data, we also have to set $C_{po} = 0.56$, as mentioned earlier. However, before carrying out the RTA and CBE analyses, it was necessary in this case to correct for a degenerate interface layer, having $\mu_{H2} = 55$ cm$^2$/V$\cdot$ s and $n_{Hsq2}/d = 3.9 \times 10^{17}$ cm$^{-3}$. The corrected data, also shown in Figures 5 and 6, are calculated by inverting Equations 26 and 27:

$$\mu_{H1} = \frac{\mu_H^2 n_H - \mu_{H2}^2 n_{Hsq2}/d}{\mu_H n_H - \mu_{H2} n_{Hsq2}/d} \tag{32}$$

$$n_{H1} = \frac{(\mu_H n_H - \mu_{H2} n_{Hsq2}/d)^2}{\mu_H^2 n_H - \mu_{H2}^2 n_{Hsq2}/d} \tag{33}$$

The fitted parameters are $N_D = 2.1 \times 10^{17}$ cm$^{-3}$, $N_A = 5 \times 10^{16}$ cm$^{-3}$, and $E_D = 16$ meV.

## SAMPLE PREPARATION

In deciding what kind of geometrical structure to use for a particular application, several factors should be considered, including (1) available size, (2) available fabrication techniques, (3) limitations of measurement time, (4) necessary accuracy, and (5) need for magnetoresistance data.

In considering these factors, we will refer to Figure 7, which depicts six of the most popular structures.

The available size can be a severe constraint. In our laboratory, we have often measured bulk samples of dimension 2 mm or less, which virtually rules out any complicated shapes, such as (B), (D), or (F) in the figure; in fact, it is sometimes not possible to modify an existing sample shape at all. Thus, the best procedure for small samples

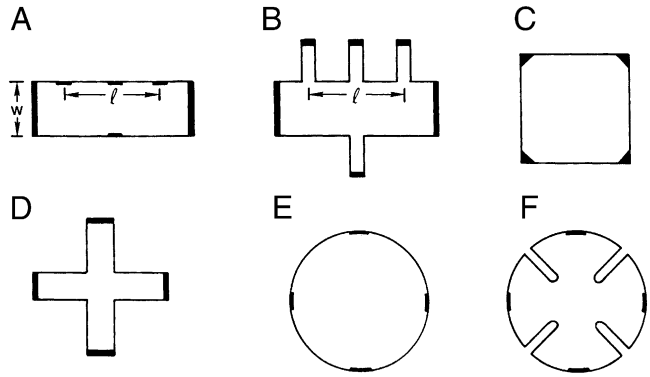



**Figure 7.** Various specimen/contact patterns commonly used for resistivity and Hall effect measurements: (**A**) Hall bar; (**B**) Hall bar with contact arms; (**C**) square; (**D**) Greek cross; (**E**) circle; (**F**) cloverleaf.

is simply to put four very small contacts around the periphery and apply the standard van der Pauw analysis. We have found that indium, applied with a soldering iron, works well for almost any semiconductor material. Contact size errors can be estimated if the shape is somewhat symmetrical. The simple Hall bar structure (A) is not recommended for a very small bulk sample unless the sample is already in that form, because it is then necessary to measure $l$ and $w$, which can introduce large errors.

For larger bulk samples, there is a greater choice among the various structures. The availability of an ultrasonic cutting tool opens up the possibility of using structures (B), (D), (E), or (F), if desired. Here, it might be noted that (B) is rather fragile compared to the others. If the samples must be cleaved from a wafer, then the shapes are basically limited to (A) and (C). In our laboratory, for example, it is common to use a square (C) of about 6 × 6 mm and then put contacts of dimension 1 mm or less on the corners.

If photolithographic capabilities are available, then one of the more complex test structures (B), (D), or (F) should be used, because of the advantages they offer. Most manufacturers of discrete semiconductor devices or circuits include Hall bar or van der Pauw structures at several points on the wafer, sometimes one in every reticle (repeated unit).

Another possible constraint is measurement time. By comparing Equation 13 with Equation 17 and Equation 14 with Equation 24, it appears that a typical van der Pauw experiment should take about twice as long as a Hall bar experiment, and indeed, this is experimentally the case. Thus, in an experiment that involves many measurements on the same sample, such as a temperature dependence study, it may well be a distinct advantage to use a Hall bar instead of a van der Pauw pattern. Also, if a contact-switching network (Fig. 4) is not part of the available apparatus, then a van der Pauw structure cannot be conveniently used.

If high accuracy is necessary for the Hall effect measurements, then structures (B), (D), and (F) are the best, because contact size effects are much stronger for Hall effect data than for resistivity data. The same structures,

along with (A), should be used if $\mu B$ must be made large, since structures (C) and (E) do not have as good a $V_H$-vs.-$B$ linearity. From a heat dissipation point of view, structures (A) and (B) are the worst, since they are required to be long and narrow, and thus the Hall voltage is relatively small for a given current, since $V_H \propto w$. A high heat dissipation can lead to large temperature gradients, and thus stronger thermomagnetic effects, or may simply raise the sample temperature an unacceptable amount.

Finally, it is important to ask whether or not the same sample is going to be used for magnetoresistance measurements, because, if so, structures (A) and (B) are undoubtedly the best. The reason is that the analysis of magnetoresistance is more complicated in van der Pauw structures than in Hall bar structures, and, in general, a simple formula cannot be found. It may be noted that in Shubnikov–de Haas and quantum Hall measurements, in which magnetic field dependence is critical, the Hall bar is nearly always used.

## PROBLEMS

### Contact Size and Placement Effects

Much has been written about this subject over the past few decades. Indeed, it is possible to calculate errors due to contact size and placement for any of the structures shown in Figure 7. For (A), (C), and (E), great care is necessary, while for (B), (D), and (F), large or misplaced contacts are not nearly as much of a problem. In general, a good rule of thumb is to keep contact size and distance from the periphery each below 10% of the smallest sample edge dimension. For Hall bar structures (A) and (B), in which the contacts cover the ends, the ratio $l/w > 3$ should be maintained.

### Thermomagnetic Errors

Temperature gradients can set up spurious electromotive forces that can modify the measured Hall voltage. Most of these effects, as well as misalignment of the Hall contacts in structure (B), can be averaged out by taking measurements at positive and negative values of both current and magnetic field and then applying Equations 21 and 25.

### Conductive Substrates

If a thin film is grown on a conductive substrate, the substrate conductance may overwhelm the film conductance. If so, and if $\mu_{sub}$ and $n_{sub}$ are known, then Equations 32 and 33 can be applied, where layer 2 is the substrate. If the substrate and film are of different types (say, a $p$-type film on an $n$-type substrate), then a current barrier ($p$-$n$ junction) will be set up, and the measurement can possibly be made with no correction. However, in this case, the contacts must not overlap both layers.

### Depletion Effects in Thin Films

Surface states as well as film-substrate interface states can deplete a thin film of a significant fraction of its charge carriers. Suppose these states lead to surface and interface potentials $\phi_s$ and $\phi_i$, respectively. Then regions of width $w_s$ and $w_i$ will be depleted of their free carriers, where

$$w_{s(i)} = \left[\frac{2\varepsilon_0 \phi_{s(i)}}{e(N_D - N_A)}\right]^{1/2} \tag{34}$$

Here it is assumed that $\phi_{(s)i} \gg kT/e$ and $e\phi_{(s)i} \gg \varepsilon_C - \varepsilon_F$. The electrical thickness of the film will then be given by $d_{elec} = d - w_s - w_i$. Typical values of $\phi_s$ and $\phi_i$ are 1 V, so that if $N_D - N_A = 10^{17}$ cm$^{-3}$, then $w_s + w_i \approx 2000$ Å $=$ 0.2 μm in GaN. Thus, if $d \approx 0.5$ μm, 40% of the electrons will be lost to surface and interface states, and $d_{elec} \approx$ 0.3 μm.

### Inhomogeneity

A sample that is inhomogeneous in depth must be analyzed according to Equations 11 and 12. In simple cases (e.g., for two layers) such an analysis is sometimes possible, as was illustrated in Figures 5 and 6. However, if a sample is laterally inhomogeneous, it is nearly always impossible to carry out an accurate analysis. One indication of such inhomogeneity is a resistivity ratio $Q \gg 1$ (Fig. 3) in a symmetric sample, which would be expected to have $Q = 1$. The reader should be warned to never attempt an $f$ correction (Fig. 3) in such a case, because the $f$ correction is valid only for sample shape asymmetry, not inhomogeneity.

### Nonohmic Contacts

In general, high contact resistances are not a severe problem as long as enough current can be passed to get measurable values of $V_c$ and $V_H$. The reason is that the voltage measurement contacts carry very little current. However, in some cases, the contacts may set up a $p$-$n$ junction and significantly distort the current flow. This situation falls under the "inhomogeneity" category, discussed above. Usually, contacts this bad show variations with current magnitude and polarity; thus, for the most reliable Hall measurements, it is a good idea to make sure the values are invariant with respect to the magnitudes and polarities of both current and magnetic field.

## ACKNOWLEDGMENTS

The author would like to thank his many colleagues at Wright State University and Wright-Patterson Air Force Base who have contributed to his understanding of the Hall effect and semiconductor physics over the years. He would also like to thank Nalda Blair for help with the manuscript preparation. Finally, he gratefully acknowledges the support received from the U.S. Air Force under Contract F33615-95-C-1619.

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